INCH-POUND

MIL-C-24640/21C
26 June 1995
SUPERSEDING
MIL-C-24640/21B

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, FOUR CONDUCTOR, POWER, TYPE FXW (INCLUDING VARIATION TYPES FXOW AND FXWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION (Watertight)

FIRST - Copper conductor, uncoated in accordance with ASTM B-8, 7 strands (watertight). AWG size and dimensions in accordance with tables I, II or III for the applicable type.

SECOND - Mica glass tape insulation 0.0029 ±.0002 inch (50 percent overlap maximum and minimum overlap of 45 percent).

THIRD - Opaque-pigmented polyimide tape .001 inch (50 percent overlap minimum).

FOURTH - Opaque-pigmented polyimide tape .001 inch with 0.0001 inch Fluorinated Ethylene Propylene [FEP] (50 percent overlap minimum).

FIFTH - Modified aromatic polyimide resin coating 0.0005 inch minimum (.001 inch nominal). Identification codes applied using Method 1 or 3.

SIXTH - Four component wires shall be cabled together. Fillers may be used as necessary, to form a round cable.

SEVENTH - Binder tape applied helically with overlap.

AMSC N/A FSC 6145

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CONSTRUCTION VARIANT FOR TYPE FXW

EIGHTH - Crosslinked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE FXWA

EIGHTH - Crosslinked polyolefin jacket. (See Table II for thickness).

NINTH - Braided metal armor. Dimension in accordance with Table II.

CONSTRUCTION VARIANT FOR TYPE FXOW

EIGHTH - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.

NINTH - Binder tape applied helically with overlap.

TENTH - Crosslinked polyolefin jacket. (See Table IV for thickness).

TABLE I. Construction details - unarmored.

Pin M24640/20	Type Number	AWG size	Conductor diameter (inch) (nominal)	Component wire diameter (inch) (min)	Component wire diameter (inch) (max)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01UN	FXW-3	16	0.058	0.075	0.081	0.034	0.266	0.286	78.2	4.35
-02UN	FXW-4	14	.073	.091	.097	.038	.315	.339	115.1	2.88

TABLE II. Construction details - armored.

Pin M24640/20	Type Number	AWG size	Conductor diameter (inch) (nominal)	Component wire diameter (inch) (min)	Component wire diameter (inch) (max)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01AN	FXWA-3	16	0.058	0.075	0.081	0.034	0.316	0.336	103.9	4.35
-02AN	FXWA-4	14	.073	.091	.097	.038	.365	.389	141.8	2.88

TABLE III. Construction details - overall shield.

Pin M24640/20	Type Number	AWG size	Conductor diameter (inch) (nominal)	Component wire diameter (inch) (min)	Component wire diameter (inch) (max)	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01UO	FXOW-3	16	0.058	0.075	0.081	0.240	0.038	0.324	0.350	131.6	4.35
-02UO	FXOW-4	14	.073	.091	.097	.262	.041	.366	.394	164.9	2.88

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REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

Component wire:

For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Blocking: Oven temperature, 200± 2°C (392± 3.6°F) no sticking of conductors
Flammability: 3 seconds (max) after flame.
3.0 inches (max) flame travel.
No flaming of tissue paper.
Humidity resistance: 5 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test: 100 percent test. Impulse voltage at 8.0 kV.
Insulation resistance: 2500 megohms for 1000 feet (min).
Polyimide cure test: No cracking.
Shrinkage: 0.062 inch maximum at 180°C
Surface resistance: 5 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F).
Maximum change in measurement: 0.031 inch.
Wet dielectric test: 2500 volts (rms).

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	See tables I, II and III
* Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield (FXOW only)	1000
Shield to shield	Not applicable
* Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	2500
Conductor to overall shield (type FXOW only)	2500
Shield to shield (component shield only)	Not applicable
Shield to shield (component shield to overall shield	Not applicable
Conductor and shield continuity	Not applicable
Jacket flaws (type FXOW only)	No failure

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	Group A:	
	Visual and dimensional	No failure
	Capacitance:	
	Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
	Capacitance unbalance - percent, maximum	Not applicable
	Characteristic impedance - at 1 MHz, ohms	Not applicable
	Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
	Weight	No failure
	Group B:	
	Cold bending	No failure
	Drip	No failure
*	Cross link proof test (percent maximum)	
	Jacket	50
	Tear - pounds per inch thickness, minimum	35
	Physicals (unaged)	
	Jacket	
	Tensile strength - lb/in^2, minimum	1300
	Elongation - percent, minimum	160
	Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
	Gas flame (1 hour)	No Failure
	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged, minimum	50
*	Elongation - percent of unaged minimum	50
*	Permanence of printing (conductor) - Method 1 only cycles, minimum	25
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - conformance to material, (type FXWA only) construction and coverage	No failure
	Shield - conformance to material, construction and coverage (type FXOW only)	No failure
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	No failure
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
*	Surface transfer impedance (type FXOW only)	
	Milliohms per meter, maximum	700
	EMP response, dB, minimum	60
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable

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Group D

	Flame propagation	No Failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Aging and compatibility (cable)	No failure
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Immersion	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged, minimum	50
*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
*	Gas Flame (3 hour)	No failure
	Component wire	No failure

Unit ordering length: 500 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review Activities:
Army - AV, CR, ER, ME, AR, AL
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DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

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